Darshil Kotecha

AI/ML Developer

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Summary

With a strong interest in AI, I have developed deep learning, computer vision, and data analytics skills through internships. Python and AI toolkit proficient, I have deployed and trained models with NVIDIA TAO Toolkit and multiple versions of YOLO, with experience in data annotation and real-time deployment. My expertise extends to NVIDIA DeepStream SDK, Apache Kafka, and PostgreSQL to scalable AI-driven video analytics and messaging systems. I prioritize ongoing development and translating research into practical AI applications for driving innovation.

Skills

Languages & Databases: Python, MySQL, PostgreSQL
Frameworks/Libraries: Scikit-learn, PyTorch, TensorFlow, Nvidia DeepStream, Nvidia TAO, Memgraph
Developer Tools: Git, GitHub, Docker/Podman, PowerBI, Voxel FiftyOne, Apache Kafka, Nvidia NGC
Soft Skills: Communication, Teamwork, Problem-Solving, Adaptability, Time Management, Leadership

Education

Symbiosis International University Pune, India
B.Tech in Artificial Intelligence and Machine Learning *2021 - 2025*

Work Experience

AI4M Technology Pvt Ltd October 2024 - June 2025
SDE Deep Learning Intern

- Trained deep learning object detection and classification models using ResNet and YOLO (v7, v9, Ultralytics v11), from data preparation to evaluation.
- Built scalable video analytics AI inference pipelines on top of NVIDIA DeepStream, with Kafka for messaging and PostgreSQL for data storage.
- Used Python and Voxel51 for automated data processing and visualization and Memgraph for relationship analysis to enhance model performance and interpretability.

Projects

BERT Model Development - Gujarati Language
Designed a domain-specific BERT model for contextual understanding in Gujarati NLP tasks.

- Developed and optimized a BERT transformer model specifically designed for the Gujarati language, concentrating on Next Sentence Prediction (NSP) tasks to enhance language comprehension and contextual modeling.

Empowering Authenticity: GAN-Based Image Forgery Detection
Developed a culturally relevant forgery detection system using GANs trained on rare Gujarati manuscript images.

- Developed a Generative Adversarial Network (GAN) model for detecting fake and modified photos, assuring visual content integrity despite advanced editing tools.

Reinforcement Learning-Based Gujarati Duha Poetry Game
Developed the game world based on MDP and used Q-Learning to enable adaptive game play.

- Developed an interactive game in which players complete Gujarati Duha stanzas by picking missing words, allowing the system to learn and improve from player interactions .